

Annexure-VI: Detailed Assessment Criteria

Detailed assessment criteria for each NOS/Module are as follows:

NOS/Module Name	Assessment Criteria for Performance Criteria/Learning Outcomes	Theory Marks	Practical Marks
NOS 1: Perform basic calculation using spreadsheet	Features of Spreadsheet Creating and saving worksheet and workbook Layouts, text formats, alignment Basic functions and formulas, sorting Graphs	23.52	10
NOS 2: Manage structured data	Introduction to database , advantages of DBMS, Concept of keys: candidate key, primary key, alternate key, Foreign key, Fundamental integrity rules: entity integrity, referential integrity. Entity-relationship model .ER-diagram, SQL, Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations. Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations.	23.52	10
NOS 3: Analyse data using spreadsheet tool	Data, table & formatting Conditional formatting Advanced charting Pivot Table Data Validation Filtering What if analysis, Role of Maths and Statistical techniques in Data Science, Probability and Statistics, Regression analysis, Descriptive	35.29	15

	statistics, Linear Regression		
NOS 4: Manage data in Open source tool	Introduction to Virtual Machine, creating and configuring Virtual Machine, Linux History, Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu, Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics, Browsing the File System, Understanding File System Concept, Managing Files, Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing Text Files, Using a Command Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files.	17.64	7.5
NOS 5: Visualize data graphically	Introduction Tableau , Connecting to Excel, CSV Text Files, Connecting to Databases, Analyzing, Formatting, Sorting and charts	31.57	15
NOS 6: Installation of Hadoop Framework and Java Programming	Installation of Hadoop, Java concepts, OOPS concepts, Looping techniques, Methods, Method Overloading, Method Over-riding, Arrays, Java Programming, HDFS, Architecture of Hadoop, Configuring Hadoop Cluster & Hadoop Commands.	36.84	17.5
NOS 7: Manage big data using hadoop	Big Data Concepts, Need for analyzing Big Data, its roles in Business Intelligence and decision making. Analysis of Data Using Mapreduce and Hive. Implementation with examples	15.78	7.5
NOS 8: Implementation of use cases of Data Analytics	Demonstration of Project	15.78	7.5
Employability Skills		30	
OJT		30	
Total		350	

Add Topic in Module NOS4:

Programming with Python

- Python environment setup
- Datatypes, operators, loops, functions
- Data structures (lists, dict, sets, tuples)
- Modules, packages & virtual environments
- Numpy advanced (arrays, vectorization, broadcasting)
- Pandas (dataframes, series, indexing, groupby, merging, pivot)
- Data wrangling: missing values, duplicates, outliers

Add Topic in Module NOS6:

BI Tools

- Introduction to BI and Power BI
- Data Modeling Concepts
- Dashboards and ReportsDashboards and Reports
- Power BI Service and Sharing
- Connecting and Transforming Data